Cross-Modal Remote Sensing Image–Audio Retrieval With Adaptive Learning for Aligning Correlation

Jinghao Huang, Yaxiong Chen, Shengwu Xiong, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— An important challenge that existing work has yet to address is the relatively small differences in audio representations compared with the rich content provided by remote sensing (RS) images, making it easy to overlook certain details in the images. This imbalance in information between modalities poses a challenge in maintaining consistent representations. In response to this challenge, we propose a novel cross-modal RS image–audio (RSIA) retrieval method called adaptive learning for aligning correlation (ALAC). ALAC integrates region-level learning into image annotation through a region-enhanced learning attention (RELA) module. By collaboratively suppressing features at different region levels, ALAC is able to provide a more comprehensive visual feature representation. In addition, a novel adaptive knowledge transfer (AKT) strategy has been proposed, which guides the learning process of the frontend network using aligned feature vectors. This approach allows the model to adaptively acquire alignment information during the learning process, thereby facilitating better alignment between the two modalities. Finally, to better use mutual information between different modalities, we introduce a plug-and-play result rerank module. This module optimizes the similarity matrix using retrieval mutual information between modalities as weights, significantly improving retrieval accuracy. Experimental results on four RSIA datasets demonstrate that ALAC outperforms other methods in retrieval performance. Compared with state-of-the-art methods, improvements of 1.49%, 2.25%, 4.24%, and 1.33% were, respectively, achieved by ALAC. The codes are accessible at https://github.com/huangjh98/ALAC.**

***Index Terms*— Adaptive learning, cross-modal remote sensing (RS) retrieval, knowledge transfer, mutual information, region-enhanced learning attention (RELA).**

I. Introduction

With the proliferation of remote sensing (RS) satellites, the annual volume of RS images has seen a steady rise. These images find extensive applications across various domains, including land planning [1], [2], crop pest monitoring [3], [4], meteorological observation [5], military reconnaissance [6], [7], and environmental pollution surveillance [8]. Extracting valuable information swiftly from the vast amount of RS data necessitates the application of image retrieval technology. Image retrieval aims to provide image results that best match a given query request. RS data retrieval methods can be classified into unimodal retrieval and cross-modal retrieval. Unimodal retrieval involves querying and retrieving results from the same type of data. For instance, Li et al. [9] proposed an image retrieval framework that transforms continuous image feature vectors into binary hash codes suitable for image retrieval. While unimodal retrieval can be used to study differences between images, its applicability is limited to within a single data type. In contrast, cross-modal retrieval involves mapping between different data modalities and can be applied to a wide range of applications, such as RS image–text (RSIT) retrieval and RS image–audio (RSIA) retrieval. Currently, RSIT retrieval has matured relatively well, but audio input aligns more closely with human habits in practical scenarios, even though audio processing can be slightly more complex. Therefore, RSIA retrieval has also gained significant attention from researchers [10], [11], [12], [13]. Chen et al. [11] devised a multiscale cross-modal hashing model that aggregates image and audio features at different scales, mapping them into binary representations constrained by category labels for Hamming space retrieval. Ning et al. [12] considered intramodal and cross-modal pair-

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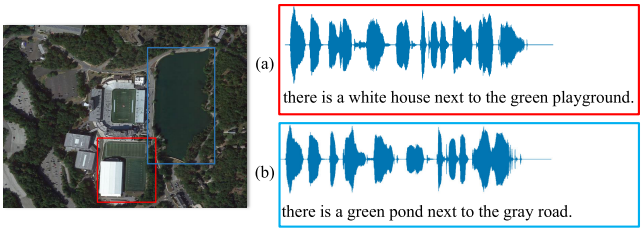


Fig. 1. Our motivational illustration. Visualization of waveform representations for different audio descriptions of the same image. (a) There is a white house next to the green playground. (b) There is a green pond next to the gray road.

wise consistency in RSIA data, effectively exploring modality features by minimizing the difference between label space and a common representation space. Huang et al. [13] introduced a multiscale transformer for extracting image features and used audio information to guide redundant removal operations within images, ultimately aligning bimodal feature vectors.

While satisfactory progress has been made in the accuracy of cross-modal RSIA methods, three significant challenges persist. First, generally speaking, RS images contain richer information compared with audio. Real-world scenes and events are captured in RS images, whereas audio provides abstract descriptions of events observed or experienced by humans. This disparity results in a notable asymmetry of information between the RS image and audio modalities. In the realm of audio, events are typically described in a less comprehensive manner, lacking thorough descriptions of event attributes and object details. In addition, differences between individuals and variations in language habits contribute to differing descriptions of the same event. And the existing RSIA retrieval methods [11], [12], [14], [15] tend to rely more on pronunciation than semantics for audio feature extraction, further exacerbating the information asymmetry issue. For example, in Fig. 1, despite significant textual disparities between Fig. 1(a) and (b), and notable differences in corresponding regions of the image, the resulting audio features lack distinctiveness. To address this issue, an innovative cross-modal retrieval method for RS images and audio, termed adaptive learning for aligning correlation (ALAC), is introduced in this study. The ALAC framework uses an adaptive knowledge transfer (AKT) strategy, using aligned feature vectors to guide the learning process of the frontend network, enabling the model to autonomously acquire alignment information and thereby facilitating improved alignment between the two modalities. Besides, this approach enhances the unique semantic features of the audio modality while reducing the influence of audio pronunciation on audio features. Second, the existing RSIA retrieval methods tend to capture prominent regions in images while neglecting other crucial areas essential for retrieval performance. To extract sufficient image information, a region-enhanced learning attention (RELA) module is ingeniously introduced, enabling comprehensive understanding of complex backgrounds through 1-D convolutional operations across various segments of RS images. The third challenge lies in the oversight of the impact of ranking results between different modalities on the similarity matrix during the inference process. The retrieval process should be mutual, meaning that if a certain RS image sample ranks among the top η candidate retrieval results, the corresponding audio sample should also rank among the top η candidate retrieval results. Achieving this reciprocity in the retrieval process is crucial for determining how to adjust the similarity matrix. Hence, a bidirectional sorting weighted fusion rerank (BSWFR) method is devised, optimizing the similarity matrix using intermodal retrieval mutual information to effectively enhance retrieval accuracy. The experimental results on four RSIA datasets demonstrate that ALAC outperforms other methods in retrieval performance. The primary contributions of this study are as follows.

1) To better comprehend the intricate and diverse backgrounds present in RS images, we introduce an innovative RELA module. This module leverages distinct groups of 1-D convolutions to conduct region-based learning on image information. These groups exhibit mutual inhibition, facilitating our ALAC model in obtaining more comprehensive visual feature representations.
2) An AKT strategy is proposed, leveraging aligned feature vectors to guide the learning process of the frontend network. This enables the model to flexibly acquire alignment information, facilitating improved modality alignment. Concurrently, this strategy enhances the unique semantic features of the audio modality while mitigating the impact of audio pronunciation on audio features.
3) To harness the full potential of mutual information between different modalities, we develop a plug-and-play method known as the BSWFR. This method uses intermodal retrieval information as weights to optimize the similarity matrix, resulting in a significant enhancement in retrieval accuracy.

Building upon the aforementioned foundation, the remaining sections are organized as follows: Section II provides a concise overview of relevant research works in relation to this study. In Section III, we present a detailed exposition of our proposed method. Section IV compares our approach with the state-of-the-art methods, presenting extensive ablation, parameter experiments, and visualized retrieval results to assess the effectiveness of our approach. Finally, Section V summarizes the entire article.

II. Related Work

In this section, we provide a concise overview of relevant literature across three key areas: rereanking strategy, salient region learning, and audio discriminative learning.

A. Reranking Strategy

Reranking technology has garnered significant attention in various retrieval tasks, including person reidentification [16], [17], [18], [19], object retrieval [20], [21], [22], and text-based image search [23], [24], [25], with the primary aim of enhancing retrieval result accuracy. In these tasks, the initial ranking lists can be further optimized by reordering the order of retrieval candidates. For instance, researchers such as Leng et al. [17] proposed a bidirectional ranking approach, which calculates new similarity scores by considering the contextual similarity between query images. Meanwhile, Zhong et al. [26] developed a novel feature vector

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for a given image after the initial ranking and performed reordering based on Jaccard distance. On a different note, Yang and Hanjalic [24] introduced the concept of supervised learning for visual search reranking, thereby improving performance through the application of reranking features relevant to the query. In traditional retrieval tasks, when using different modalities to query samples, the generated similarity matrices may result in different ranking of results between different modalities. To address this issue, Wang et al. [27] and Yuan et al. [28] proposed cross-modal reranking algorithms, aiming to adjust the retrieval result ranking between different modalities to better reflect their mutual importance. However, these methods do not take into account the fact that as the number of retrievals becomes larger, the influence of forward and reverse becomes smaller and smaller, and then degenerates back to the traditional retrieval method.

B. Salient Region Learning

In recent years, there has been significant research interest in the domain of fine-grained image processing, as demonstrated by several studies [29], [30], [31], [32]. Unlike general image processing, fine-grained processing delves deeper into image attributes such as color, shape, and texture. This is particularly relevant in the context of image retrieval, where fine-grained processing poses specific challenges. There are two main categories within fine-grained image retrieval: content-based fine-grained image retrieval and sketch-based fine-grained image retrieval. An early example of content-based fine-grained image retrieval is presented by sCDA [33]. This approach uses an unsupervised pretrained model for feature extraction and information localization, followed by cascading through max-pooling and average-pooling operations. Scharfenberger et al. [34] used a multilayer approach to analyze the structural and textural characteristics of natural images as important features for salient region detection from a scale point of view. Zhou et al. [35] proposed a bottom-up salient region detection method that integrates compactness and local contrast cues. On the other hand, sketch-based fine-grained image retrieval involves matching hand-drawn sketches to corresponding image instances. The existing techniques in this area aim to bring sketches and images closer in a shared embedding space [32], [36]. However, much of the prior work has focused on natural images. RS images face unique challenges primarily due to their characteristic wide-angle perspective capturing scenes, resulting in subtle variations in image content. This extensive field of view makes the identification of salient features quite challenging when determining image features mapped to image vectors. Consequently, addressing the issue of saliency estimation remains a challenging endeavor within the realm of fine-grained image retrieval, especially in the context of RS images.

C. Audio Discriminative Learning

Salamon and Bello [37] explored the use of deep convolutional neural networks and data augmentation in the task of environmental sound classification to enhance the discriminative features of audio. Salamon and Bello [37] focused on data feature selection, using methods such as correlation coefficients and mutual information to streamline audio features and improve model performance. Luo et al. [38] delved into the application of deep clustering methods in audio separation, involving clustering of audio features to enhance differentiation between different sources. This is crucial for both audio separation and source identification. Lung et al. [39] introduced a method for end-to-end speaker verification using raw waveforms. Handling the raw waveforms directly contributes to enhancing the discriminative features of speakers. Yang et al. [40] proposed various approaches for fusing audio and text information to improve the recognition capabilities of audio patterns. The aforementioned methods primarily focus on distinguishing features of different sources in audio or guiding feature learning by introducing additional information. However, these methods have not achieved significant differentiation at the semantic information level in audio.

III. Method

To address the challenge of cross-modal retrieval between an RS image and a segment of audio, we present the architecture of the deep RSIA network as depicted in Fig. 2. The network comprises five key components: problem definition, audio representation, image representation, BSWFR, and objective function.

A. Problem Definition

Given N paired units $\mathcal{U} = \{v, a\}$, where v represents an RS image and a represents a segment of audio, our ALAC method aims to use deep networks to learn feature representation $\boldsymbol{f}_v'$ for an RS image and $\boldsymbol{f}_a'$ for a segment of audio. The task is to learn a suitable projection function by leveraging a multimodal training dataset and the provided paired unit $\mathcal{U}$. This function ensures the preservation of similarity between RS image and audio.

B. Audio Representation

According to the description in [41], the audio segments are initially sampled at a rate of 16 kHz. Subsequently, they undergo processing through a short-time Fourier transform (STFT) with a window size of 1024 and a hop length of 200. The resulting spectrogram is mapped to the Mel-frequency cepstral coefficients scale, generating an audio matrix $M^{T\times F}$ with $T = 400$ and $F = 64$. Our audio branch uses a ResNet18 network as the audio feature extractor Ψ, consistent with the approach of Huang et al. [13], for modeling the audio content. Finally, the output of the fully connected layer serves as the audio representation $\boldsymbol{f}_a \in \mathbb{R}^{d_f}$

$$\boldsymbol{f}_a = \Psi(a, \theta_a) \tag{1}$$

where $\boldsymbol{f}_a$ represents the audio representation obtained through the function Ψ, and θ_a is the parameter used to represent the characteristics of Ψ.

C. Image Representation

Compared with natural images, RS images exhibit variable scales and orientations. RS images typically contain rich information about objects. While the existing RSIA methods aid in extracting RS information, they tend to focus on capturing prominent regions within the images. This inclination may

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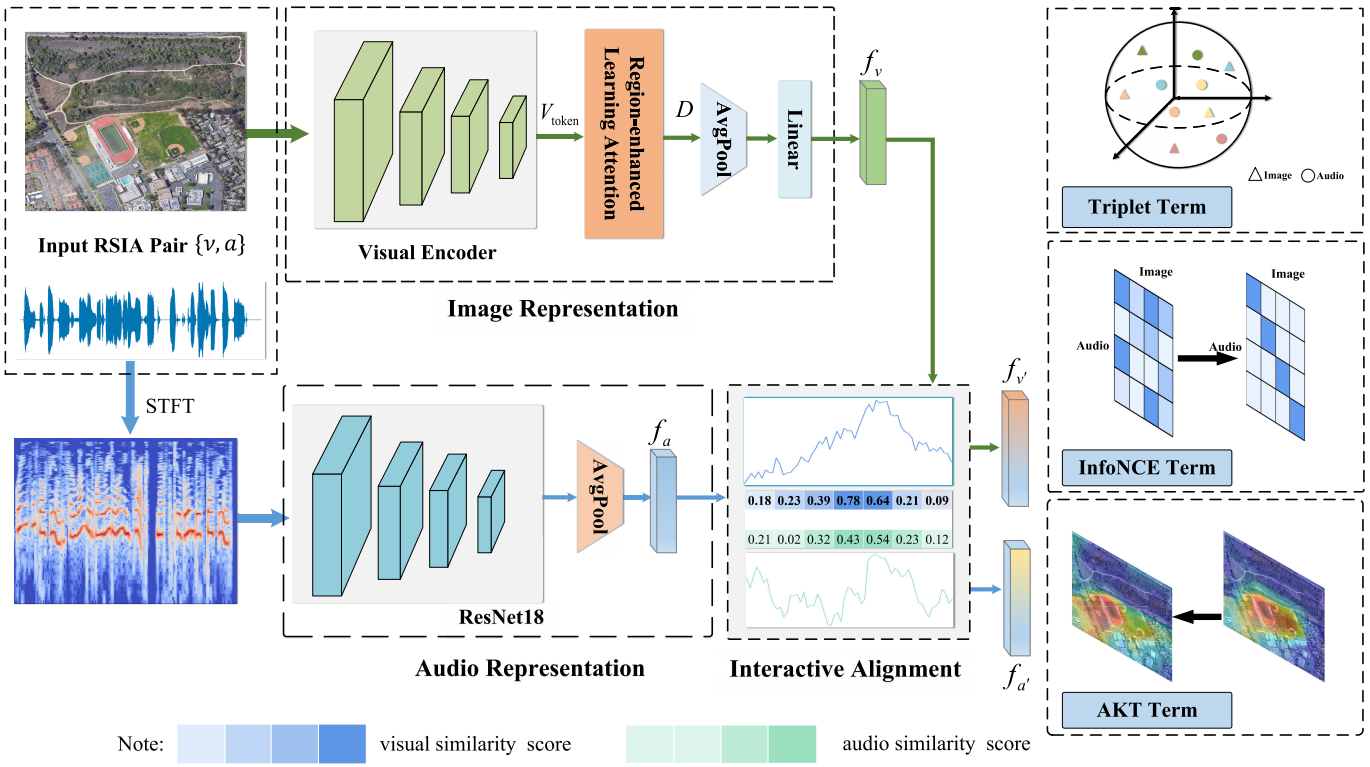


Fig. 2. ALAC framework takes RSIA pairs as input and comprises three main components: the RS image branch, the audio branch, and the IA module. The RS image branch includes a visual encoder, an RELA module, an average pooling layer, and a fully connected layer. The audio branch consists of a ResNet18 network and an average pooling layer. The overall objective function is composed of three terms: the triplet term, the InfoNCE term, and the adaptive knowledge transfer term.

lead to the neglect of less salient but crucial areas for retrieval performance. To solve the above problems, we construct an RELA module.

1) Visual Encoder: To obtain features from images, Huang et al. [13] proposed a transformer Φ with a multiscale fusion module. Therefore, we use Φ to extract image features, which can be represented as

$$\boldsymbol{V}_{(\textbf{token})} = \Phi(v, \theta_v) \in \mathbb{R}^{L \times d_k} \tag{2}$$

where $\boldsymbol{V}_{(\textbf{token})}$ is the image feature maps extracted using the Φ, θ_v denotes the parameters of Φ, d_k indicates the dimension of visual features, and L represents the sequence length.

2) Region-Enhanced Learning Attention: In this section, an RELA is proposed. It is shown in Fig. 3. Specifically, for the input image information $\boldsymbol{V}_{(\textbf{token})}$, we use different region learning groups to extract features from distinct regions. We have defined n region learning groups, which can be represented as follows:

$$\boldsymbol{O}'_i = \begin{cases} \mathcal{C}_1\left(\boldsymbol{V}_{(\textbf{token})}\right), & i = 1 \\ \mathcal{C}_i\left(\delta(\varphi(\boldsymbol{O}_{i-1})) \otimes \boldsymbol{O}'_{i-1}\right), & i \in [2, n] \end{cases} \tag{3}$$

where $\boldsymbol{O}'_i$ denotes the visual features extracted by each region learning group. $\mathcal{C}(x)$ represents a 1-D convolution operation. φ represents the softmax function. δ represents a standardization operation. $\otimes$ represents the dot product. $\boldsymbol{O}_i (i \in [2, n])$ represents the visual features obtained after suppressing the learned visual regions from the previous region learning group.

To provide more details, the first region learning group is initialized using $\boldsymbol{O}_1$ from the first stage, and then the subsequent $n-1$ region learning groups are computationally

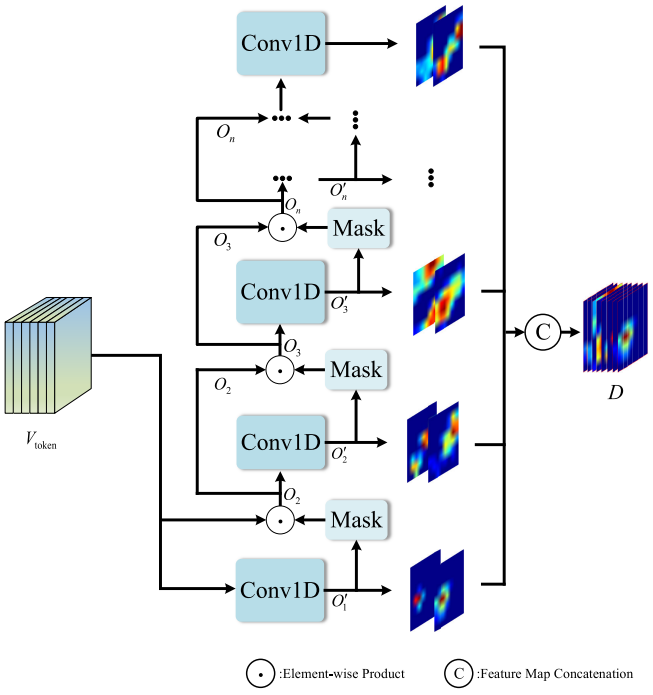


Fig. 3. Elaborate on the RELA.

generated using $\boldsymbol{O}_i (i \in [2, n])$. To obtain $\boldsymbol{O}'_1$, we directly encode $\boldsymbol{V}_{(\textbf{token})}$ using $\mathcal{C}_1$ to obtain $\boldsymbol{O}'_1$. The aforementioned operation is as follows:

$$\boldsymbol{O}'_1 = \mathcal{C}_1\left(\boldsymbol{V}_{(\textbf{token})}\right) \in \mathbb{R}^{L \times d_k}. \tag{4}$$

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Then $\boldsymbol{O}_1'$ goes through the softmax function to compute the weight of each region, followed by a normalization operation to acquire the attention toward the regions that should be suppressed in learning. Then, the latest image features are obtained by calculating the correlation between this attention and the image features, as shown below

$$\boldsymbol{O_2} = \left(2 - \left(\frac{\varphi(\boldsymbol{O}_1') - \mu_1^{\text{mean}}}{\mu_1^{\text{std}}} + 1\right)\right) \otimes \boldsymbol{V}_{(\textbf{token})} \tag{5}$$

where φ represents the softmax function, and μ_1^{mean} and μ_1^{std} denote the mean and standard deviation of $\varphi(\boldsymbol{O}_1')$, respectively. The remaining region learning groups are generated in a cyclical manner using operations that involve establishing relationships between each region, inhibiting the learning of regions already learned by previous groups, and enhancing other activated regions through 1-D convolution. The operations for the remaining region learning groups are detailed as follows:

$$\boldsymbol{O_i'} = \mathcal{C}_i(\boldsymbol{O_i}) \in \mathbb{R}^{L \times d_k}, \quad i \in [2, n]. \tag{6}$$

Then $\boldsymbol{O}_{\boldsymbol{i-1}}'$ calculates the weights of each region using the softmax function, and then undergoes a normalization operation to determine the attention toward regions to be suppressed in learning. Finally, the latest image features are obtained by calculating the correlation between this attention and the image features, as shown below

$$\boldsymbol{O_i} = \left(2 - \left(\frac{\varphi(\boldsymbol{O}_{i-1}') - \mu_{i-1}^{\text{mean}}}{\mu_{i-1}^{\text{std}}} + 1\right)\right) \otimes \boldsymbol{O_{i-1}}, \quad i \in [2, n]. \tag{7}$$

In summary, using a suppression-enhancing mask-based attention operation, we combine all $\boldsymbol{O_i'}$ to obtain the final output $\boldsymbol{D}$. This operation partially restrains the most discriminative regions from the attention guidance of the previous stage, while enhancing the activated regions. This approach allows us to maintain the correlations between the activated regions from the previous stage and those subsequently generated. After obtaining the visual features $\boldsymbol{D}$, we perform average pooling on $\boldsymbol{D}$ and use a linear transformation to generate the image representation $\boldsymbol{f_v} \in \mathbb{R}^{d_f}$, as follows:

$$\boldsymbol{f_v} = \mathcal{F}(\mathcal{A}(\boldsymbol{D})) \tag{8}$$

where $\mathcal{A}(x)$ and $\mathcal{F}(x)$ denote the spatial averaging pooling and fully connection layer, respectively. To better align the final feature representation across different modalities and enhance the underlying semantic relationship between RS image and audio, Huang et al. [13] proposed the interactive alignment (IA) module. To achieve this, we use the IA module for feature alignment

$$\boldsymbol{f_v'}, \boldsymbol{f_a'} = \mathcal{D}_{\text{IA}}(\boldsymbol{f_v}, \boldsymbol{f_a}) \tag{9}$$

where $\boldsymbol{f_v'} \in \mathbb{R}^{d_f}$ and $\boldsymbol{f_a'} \in \mathbb{R}^{d_f}$ represent the feature representation of aligned image and audio, respectively, with $\mathcal{D}_{\text{IA}}$ being the IA structural module.

D. Bidirectional Sorting Weighted Fusion Rerank

In this section, we introduce a postprocessing method (BSWFR) for retrieval, which uses the forward and backward ranking information as weights to optimize the similarity matrix. This involves calculating forward and reverse ranking results as weights to enhance retrieval accuracy. First, in the similarity matrix $\boldsymbol{S}_{\text{es}}$, a query is defined as $\boldsymbol{Q}_{\text{es}}$

$$\boldsymbol{Q}_{\text{es}} = \{s_1, s_2, \ldots, s_k\} \tag{10}$$

where e represents the query image, k denotes the top k nearest neighbor audios, and s_i represents the audio of the ith image most similar to the query image e. Simultaneously, we obtain the ranking positions $L_s = \{L_{s_1}, L_{s_2}, \ldots, L_{s_k}\}$, and then quantify the ranking position information as follows:

$$w_{\text{es}_i} = \xi \exp^{-(L_{s_i}+1)}, \quad i \in [1, k] \tag{11}$$

where $\exp(x)$ represents e^x function. w_{es_i} quantifies the sorting position in the audio retrieval process, assigning greater weight to larger audio as the sorting position advances. ξ controls the sensitivity of the ranking position quantization. Next, based on the retrieval results obtained in the previous step, a reverse search is conducted. $\boldsymbol{Q}_{s_i e}$ represents the ranking results returned by querying using s_i in the similarity matrix $\boldsymbol{S}_{\text{es}}$

$$\boldsymbol{Q}_{s_i e} = \{e_1, e_2, \ldots, e_l\}, \quad i \in [1, k] \tag{12}$$

where l represents the number of returned images, and e_l represents the first l image results returned. For the first k audio retrieval results corresponding to query image, each of the k audios can be retrieved through reverse retrieval to obtain the first l retrieved images. Subsequently, the ranking position $L_e = \{L_{e_1}, L_{e_2}, \ldots, L_{e_q}\}$ corresponding to image e is determined from these first l reverse-retrieved images, and then quantified

$$w_{s_i e} = \zeta \exp^{-(L_e+1)}, \quad i \in [1, k] \tag{13}$$

where $w_{s_i e}$ represents the quantified representation of the ranking results from the reverse retrieval. ζ controls the sensitivity of the ranking position quantization. Furthermore, we recalibrate the similarity matrix, preserving the similarity values for the first k results and assigning 0 to other positions

$$\boldsymbol{S} = \begin{cases} \boldsymbol{S}_{\textbf{es}}[e], & e \in \boldsymbol{Q}_{\textbf{es}} \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $\boldsymbol{S}$ is used to compute the confidence of the model in the source similarity and serves as the fundamental similarity indicator. The computation of the final similarity matrix $\boldsymbol{S_R}$ involves forward and backward ranking information, which is used as weights to optimize the similarity matrix. Specifically, the final reranked similarity matrix is obtained by multiplying w_{es_i} and $w_{s_i e}$ with $\boldsymbol{S}$

$$\boldsymbol{S_R} = \boldsymbol{w}_{\textbf{es}} \boldsymbol{w}_{\textbf{se}} \boldsymbol{S}. \tag{15}$$

E. Objective Function

In the training phase, we introduce bidirectional triplets [42]) to enhance the similarity between the matching features of $\boldsymbol{f_v'}$ and $\boldsymbol{f_a'}$, while incorporating an InfoNCE term [43] to learn the prior similarity between the

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representation of RS images $\boldsymbol{f_v}$ and audios $\boldsymbol{f_a}$. Building upon this foundation, the AKT term is designed to minimize the distance between their similarity matrices. Using aligned feature vectors to guide the learning process of the front-end network, the adaptive knowledge transfer term strengthens the semantic features unique to the audio modality, while reducing the impact of pronunciation variations on audio features.

1) Bidirectional Triplet Term: The objective of the bidirectional triplet term is to learn appropriate projection functions that bring image–audio pairs with the same semantic meaning closer together in the projection space, thereby preserving the similarity between different modalities. To achieve this goal, we introduce the bidirectional triplet term $\mathcal{L}_\beta$, with its computation formula as follows:

$$\mathcal{L}_\beta = \sum_{i=1}^{N}\sum_{j=1}^{N}\left[\alpha - S\left(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_i'}\right) + S\left(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'}\right)\right]_+ + \sum_{j=1}^{N}\sum_{i=1}^{N}\left[\alpha - S\left(\boldsymbol{f}_{\boldsymbol{v}_j'}, \boldsymbol{f}_{\boldsymbol{a}_j'}\right) + S\left(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'}\right)\right]_+ \quad (16)$$

where $i = j$, $\boldsymbol{f}_{\boldsymbol{v}_i'}$ and $\boldsymbol{f}_{\boldsymbol{a}_j'}$ denote positive pairs, and when $i \neq j$, $\boldsymbol{f}_{\boldsymbol{v}_i'}$ and $\boldsymbol{f}_{\boldsymbol{a}_j'}$ denote negative pairs. α represents the margin parameter. $[x]_+ = \max(x, 0)$, and $S(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_i'})$ denotes the similarity between $\boldsymbol{f}_{\boldsymbol{v}_i'}$ and $\boldsymbol{f}_{\boldsymbol{a}_i'}$.

2) InfoNCE Term: To capture the similarity between $\boldsymbol{f}_{\boldsymbol{v}_i}$ and $\boldsymbol{f}_{\boldsymbol{a}_j}$, we introduce the InfoNCE term for contrastive learning, aiming to enhance their similarity and maximize the lower bound of mutual information between variable pairs. The computation of the InfoNCE term $\mathcal{L}_\varphi$ is as follows:

$$\mathcal{L}_\varphi = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(S\left(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_i}\right)/\tau\right)}{\sum_{j=1}^{N}\exp\left(S\left(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j}\right)/\tau\right)} \quad (17)$$

where τ represents a temperature parameter.

3) Adaptive Knowledge Transfer Term: To feed the information of the aligned retrieved vectors into the feature vectors of the output of the image and audio models, we design a distance between the similarity matrices of minimizing the two, which is calculated as

$$\mathcal{L}_\varrho = \sum_{i=1}^{N}\sum_{j=1}^{N} H_\kappa\left(S\left(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'}\right), S\left(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j}\right)\right) \quad (18)$$

where H_κ denotes the Huber loss [44] and the formula is defined as follows:

$$H_\kappa(\boldsymbol{a}, \boldsymbol{b}) = \begin{cases} \frac{1}{2}(\boldsymbol{a} - \boldsymbol{b})^2, & |\boldsymbol{a} - \boldsymbol{b}| \leq \kappa \\ \kappa\left(|\boldsymbol{a} - \boldsymbol{b}| - \frac{1}{2}\kappa\right), & \text{otherwise} \end{cases} \quad (19)$$

where κ denotes a hyperparameter. In the relationship $S(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j})$, there exist certain incorrect associations. After the alignment operation, $S(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'})$ encompasses additional information that is not perceived by $S(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j})$. Therefore, attempting to directly use all the information from $S(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'})$ to supervise the training of $S(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j})$ might confuse $S(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j})$ and could even lead to a performance decline. Consequently, in light of this scenario, we propose an adaptive distillation strategy that enables $S(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j})$ to selectively acquire relational knowledge from $S(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'})$. The constraints of this strategy serve to enhance the positive knowledge transfer and suppress erroneous correlations during the training process.

To enable the model to learn the required knowledge of the relationships between modalities in an adaptive manner, we set a dynamic weight mask matrix for the similarity matrix as follows:

$$\mathcal{L}_\varrho' = \sum_{j=1}^{N}\sum_{i=1}^{N}\boldsymbol{\omega}(i, j) H_\kappa\left(S\left(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'}\right), S\left(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j}\right)\right) \quad (20)$$

where $\boldsymbol{\omega}$ is a mask matrix whose elements are between 0 and 1. Subject to the neural structure search algorithm, $\boldsymbol{\omega}$ can be considered as an optimized structural parameter. θ denotes the model parameters. Because the search space of the model is continuous, we use the expectation–maximization (EM) algorithm [45] for iterative optimization of $\boldsymbol{\omega}$ and θ. We use $\boldsymbol{\omega}$ in the training set to update the similarity matrix and the validation set to update our $\boldsymbol{\omega}$ matrix. At the same time, to better optimize $\boldsymbol{\omega}$ and reduce the influence of similarity on the elements on the corner line, we further weighted $\boldsymbol{\omega}$ in $\mathcal{L}_\omega$ as follows:

$$\mathcal{L}_\omega(\theta, \omega) = \sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\boldsymbol{\omega}(i, j) H_\kappa\left(S\left(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'}\right), S\left(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j}\right)\right)}{S\left(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'}\right)} \quad (21)$$

where $\mathcal{L}_\omega$ indicates that $\boldsymbol{\omega}$ is updated using the validation set after freezing of θ. We randomly select samples from each training batch, and based on the symmetry of the exchange, we can use two values to represent the diagonal and nondiagonal elements of $\boldsymbol{\omega}$, respectively. When the model converges, the tendency of the adaptively learned mask matrix is to transfer the information from the diagonal of $S(\boldsymbol{f}_{\boldsymbol{v}_i'}, \boldsymbol{f}_{\boldsymbol{a}_j'})$ to $S(\boldsymbol{f}_{\boldsymbol{v}_i}, \boldsymbol{f}_{\boldsymbol{a}_j})$.

4) Total Objective Function: The overall optimization objective function $\mathcal{L}$ can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_\beta + \gamma\mathcal{L}_\varphi + \lambda\mathcal{L}_\varrho' \quad (22)$$

where γ and λ are hyperparameters used to fine-tune the performance of model. Algorithm 1 shows the process of our AKT method.

IV. Experimental Results

In this section, we begin by providing a comprehensive overview of the used datasets, evaluation metrics, and experimental setup. Following that, we conduct a thorough analysis and comparison across the introduced four datasets, showcasing the advantages of our proposed method in terms of performance compared with the existing approaches. Furthermore, we delve into exploring the impact of different model structures through a variety of ablation experiments, parameter studies, BSWFR, and AKT. Finally, we present visualizations demonstrating the enhanced learning attention regions, as well as the results of our model in the retrieval task and semantic localization.

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Algorithm 1 Adaptive Knowledge Transfer

Require: Training samples, Validation samples
Require: Initial mask parameter $\boldsymbol{\omega}$ and model parameter θ
Initialize parameters $\boldsymbol{\omega}$ and θ
repeat
  // Training Phase
  for all training samples $(\mathbf{v}_i, \mathbf{a}_i)$ **do**
    $\boldsymbol{f}_v', \boldsymbol{f}_a', \boldsymbol{f}_v, \boldsymbol{f}_a = \mathrm{ALAC}(\mathbf{v}_i, \mathbf{a}_i; \theta; \boldsymbol{\omega})$
    Calculate training loss $\mathcal{L}(\boldsymbol{f}_v', \boldsymbol{f}_a', \boldsymbol{f}_v, \boldsymbol{f}_a)$
    Update θ by $\mathcal{L}$
  end for
  // Evaluation Phase
  for all validation samples $(\mathbf{v}_j, \mathbf{a}_j)$ **do**
    $\boldsymbol{f}_v', \boldsymbol{f}_a', \boldsymbol{f}_v, \boldsymbol{f}_a = \mathrm{ALAC}(\mathbf{v}_j, \mathbf{a}_j; \theta)$
    Calculate validation loss $\mathcal{L}_\omega(\boldsymbol{f}_v', \boldsymbol{f}_a', \boldsymbol{f}_v, \boldsymbol{f}_a)$
    Update $\boldsymbol{\omega}$ by $\mathcal{L}_\omega$
  end for
until Convergence

A. Dataset and Evaluation Metrics

To validate the effectiveness of our proposed approach, we conduct experiments on four commonly used RSIA datasets, namely, SYDNEY, UCM, RSITMD, and RSICD image–audio datasets. Among these, the RSICD image–audio dataset boasts the largest sample size of 10 921 images, each with dimensions 224 × 224. The RSITMD image–audio dataset comprises 4743 samples, with image dimensions of 256 × 256. The UCM image–audio dataset contains 2100 samples, sized at 256 × 256 per image. Finally, the SYDNEY image–audio dataset consists of 613 samples, with each image having dimensions of 500 × 500. Adhering to the dataset division approach of Huang et al. [13], we allocate 70% of the RSITMD image–audio dataset as the training set, 20% as the validation set, and 10% as the test set. For the SYDNEY, UCM, and RSICD image–audio datasets, we use a split of 60% for training, 20% for validation, and 20% for testing.

In the context of retrieval tasks, we use K-recall ($R@K$) as an evaluation metric, representing the percentage of queries with true matches within the top K search outcomes. We choose values of K as 1, 5, and 10 to ensure a comprehensive evaluation of results. Subsequently, we use mean $R@K$ score (mR) to holistically measure the performance of our model, providing an intuitive assessment standard.

B. Experiment Details

The proposed ALAC method is implemented using the MindSpore and PyTorch frameworks. The experiments are conducted on a workstation equipped with a GeForce RTX 3090 GPU, Intel Xeon[1] Silver 4210R CPU at 2.40-GHz X40 CPU, and 62.6-GiB RAM. Prior to conducting the experiments, a series of data augmentation operations are applied to enhance the generalization performance of the model. The images are ultimately resized to 224 × 224 pixels and input into the model. As a benchmark for comparison to other methods, the same ResNet18 network is used to extract audio features. The embedding space for both images and audios is set to 256. Equation (22) is optimized using the Adam optimizer, with the learning rate decaying by a factor of 0.7 every 20 epochs, starting at an initial learning rate of $2e^{-4}$. The batch size is set to 128, and the margin for the contrastive loss is set to 0.2. The hyperparameters γ and λ are set to 0.1. The hyperparameter κ is set to 1. ALAC is trained for 50 epochs or until the loss no longer decreased. To enhance the credibility of the experimental results, each group of experiments undergoes two identical rounds of training and validation, and the average result is obtained.

[1]Registered trademark.

C. Comparison With the SOTA Methods

To assess the effectiveness of the ALAC method, we conduct a comparative analysis against several state-of-the-art approaches. The details of the comparisons are provided below.

1) *VSE++ [46]:* VSE++ uses VGG19 to extract image features and bidirectional GRU to extract text features and uses contrastive loss for model learning.
2) *PVSE [47]:* PVSE computes multiple locally guided feature vectors for visual and text modalities individually. These local feature vectors are then combined with the global feature vectors to generate K embeddings, resulting in a final many-to-many effect. The $K = 2$ method and the $K = 1$ method of PVSE are used as control groups.
3) *PCME [48]:* PCME achieves a one-to-many relational mapping in a shared representation space by representing the feature vectors from different modalities as identical probability distributions.
4) *CLIP [49]:* CLIP is a contrast-based cross-modal pretrained model for image–text learning. It is fine-tuned specifically for RS image–audio retrieval tasks.
5) *AMFMN [50]:* AMFMN uses a multiscale and redundant information filtering structure to handle complex image features. It then aligns images and texts using a semantic alignment structure. In the comparison experiments, we evaluate AMFMN against the soft, sim, and fusion methods proposed in the article.
6) *SSIA [13]:* SSIA uses a multiscale transformer to extract image features, then audio is used to guide the images for redundancy removal operations, and finally bimodal feature vectors are aligned with each other.

We perform experiments on four widely used datasets and compute the average test results, which are presented in Tables I and II.

1) *Results on RSICD Image–Audio Dataset:* On the left side of Table I are the retrieval results of different methods on the RSICD dataset. ALAC performs well on this dataset, with an mR that is 0.49 points higher than the best-performing model that does not use the BSWFR module. After incorporating the BSWFR module to optimize the retrieval results, it is observed that the performance of the ALAC model improves by 1.00 point in terms of mR compared with when the module is not used.
2) *Results on RSITMD Image–Audio Dataset:* On the right side of Table I are the experimental results on

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TABLE I
PERFORMANCE COMPARISONS ON RSICD AND RSITMD TEST SETS FOR RETRIEVAL TASKS

Method	RSICD dataset							RSITMD dataset						
	Audio Retrieval			Image Retrieval			mR	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	1.42	3.57	6.25	1.57	7.14	12.36	5.38	3.09	10.32	17.92	1.59	6.45	11.76	8.52
PVSE K=2	0.94	2.50	3.64	0.73	3.66	7.02	3.08	1.33	4.87	9.44	1.03	5.34	9.82	5.31
PVSE K=1	0.82	1.83	2.83	0.84	3.80	6.84	2.83	1.47	6.19	10.18	0.86	4.48	9.16	5.39
CLIP	3.44	7.38	11.25	3.31	12.19	20.06	9.61	2.64	12.39	17.04	1.28	6.37	12.12	8.64
PCME	3.41	8.45	13.09	3.18	11.70	19.45	9.90	4.94	16.00	25.00	2.29	10.19	16.86	12.55
AMFMN-soft	2.94	7.13	10.91	2.63	11.52	18.76	8.98	5.01	17.33	26.03	2.80	12.51	22.18	14.31
AMFMN-sim	3.51	7.54	11.80	3.13	11.80	19.01	9.46	4.50	15.63	27.21	2.51	13.22	22.64	14.28
AMFMN-fusion	3.69	7.73	12.13	2.66	10.46	17.39	9.01	5.09	16.48	24.89	2.39	11.53	20.33	13.45
SSIA	4.16	8.23	12.94	3.83	14.55	24.03	11.29	6.12	20.13	32.74	3.86	18.03	31.12	18.67
ALAC w/o BSWFR	3.86	8.80	14.01	3.80	14.96	25.21	11.78	6.63	**22.34**	32.96	4.75	20.39	**35.79**	20.48
ALAC with BSWFR	**4.54**	**10.51**	**15.68**	**3.99**	**15.61**	**26.36**	**12.78**	**7.30**	22.23	**34.95**	**4.84**	**21.21**	35.02	**20.92**

TABLE II
PERFORMANCE COMPARISONS ON SYDNEY AND UCM TEST SETS FOR RETRIEVAL TASKS

Method	SYDNEY dataset							UCM dataset						
	Audio Retrieval			Image Retrieval			mR	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	8.73	28.84	43.37	8.94	35.98	57.41	30.55	10.63	36.59	54.13	10.65	39.63	60.37	35.33
PVSE K=2	4.76	22.22	37.83	5.29	24.76	45.56	23.40	5.48	22.22	40.08	4.62	20.33	38.51	21.87
PVSE K=1	4.23	25.92	38.10	5.56	25.71	45.93	24.24	4.68	22.78	37.78	4.22	20.21	37.57	21.21
CLIP	12.43	29.63	38.62	9.89	35.82	54.34	30.12	17.14	45.08	56.90	16.13	45.86	62.22	40.56
PCME	12.43	34.92	49.47	11.8	37.25	57.04	33.82	18.73	48.25	63.97	17.84	51.57	71.24	45.27
AMFMN-soft	11.64	31.75	44.18	9.37	39.47	57.04	32.24	16.51	49.44	64.76	16.75	52.33	72.52	45.37
AMFMN-sim	11.90	37.57	50.79	12.38	39.84	59.63	35.35	17.46	50.16	64.84	16.24	49.76	68.02	44.41
AMFMN-fusion	11.90	32.54	49.21	13.44	41.27	61.16	34.92	15.40	46.27	62.78	15.94	50.56	69.68	43.44
SSIA	15.07	36.77	49.47	13.59	43.01	63.33	36.87	18.49	50.79	67.14	18.68	53.82	71.34	46.71
ALAC w/o BSWFR	13.06	39.28	53.17	16.67	47.62	64.28	38.99	**19.04**	50.35	67.38	**19.73**	56.09	**73.47**	47.68
ALAC with BSWFR	**17.06**	**43.25**	**57.14**	**16.74**	**48.17**	**64.28**	**41.11**	18.69	**52.26**	**67.61**	19.52	**56.33**	73.40	**48.04**

the RSITMD dataset. Due to the finer granularity of the RSITMD dataset, accurate localization of relevant regions in the images is crucial. Before using the BSWFR module, the ALAC model showcases remarkable performance, surpassing the best method by 1.81 points. This underscores that our proposed RELA module, in conjunction with the AKT term, enables a more precise localization of objects described in the audio, particularly within intricate image scenarios. After incorporating the BSWFR module, the performance further improves to 20.92, demonstrating the superiority of the model.

3) *Results on UCM Image–Audio Dataset:* On the UCM dataset, the performance gain achieved by the ALAC model is particularly prominent, as shown in Table II. Although the performance improvement is not significant without the optimization of the BSWFR module, the inclusion of the BSWFR module leads to a notable improvement in the mR metric. This further confirms the robustness of the model.
4) *Results on SYDNEY Image–Audio Dataset:* The SYDNEY dataset is relatively small. Our proposed model still achieves the best performance on this dataset. As shown in Table II, the ALAC model improves the mR by 2.12 points compared with SSIA. This further validates that the RELA module alleviates the underutilization of visual information, while the AKT term enhances semantic localization among different modules. Subsequently, by leveraging the BSWFR module to optimize the retrieval results and emphasize the importance of performance improvement, the mR indicator of the ALAC model reaches its optimal level at 41.11.

D. Ablation Experiments

When comprehensively evaluating the performance of the proposed different modules of the ALAC method, a series of experiments are conducted. Initially, a baseline method (m_1) is introduced. In the first experiment, learning is solely based on the RELA module (m_2). Subsequently, in the second experiment, cross-modal retrieval is conducted using only the AKT term (m_3). The proposed ALAC method without the BSWFR module is implemented in the third experiment (m_4). Finally, the complete ALAC method is trained in the fourth experiment (m_5). From Table III, it is evident that the proposed method m_5 outperforms m_1, m_2, m_3, and m_4 in terms of performance. It is noteworthy that when using only the RELA module or the AKT term, our results show improvements of

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TABLE III
ABLATION EXPERIMENTS ARE CONDUCTED ON THE RSITMD TEST DATASET TO EVALUATE THE PERFORMANCE OF DIFFERENT STRUCTURES OF ALAC

Configurations	RELA	AKT	BSWFR	Audio Retrieval			Image Retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
m_1	×	×	×	6.52	**22.44**	33.07	4.13	16.81	28.53	18.59
m_2	✓	×	×	5.42	21.12	34.73	4.20	18.16	30.86	19.08
m_3	×	✓	×	6.41	21.12	32.85	4.53	19.49	34.15	19.76
m_4	✓	✓	×	6.63	22.34	32.96	4.75	20.39	**35.75**	20.48
m_5	✓	✓	✓	**7.30**	22.23	**34.95**	**4.84**	**21.21**	35.02	**20.92**

TABLE IV
COMPARISON EXPERIMENT OF RERANK ON THE RSITMD TEST DATASET

Reranking Method	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
Source	6.63	22.34	32.96	4.75	20.39	**35.75**	20.48
Rerank	5.19	19.91	33.93	4.29	19.60	32.96	18.83
Multivariate Rerank	6.96	**22.67**	34.95	4.66	21.11	34.53	20.81
BSWFR	**7.30**	22.23	**34.95**	**4.84**	**21.21**	35.02	**20.92**

0.49%–1.17% in the mR metric compared with the baseline method, demonstrating the performance advantages of these two modules. This suggests that the RELA module explicitly incorporates ample visual semantics, while the combination of the AKT term and RELA module significantly enhances model performance. This indicates that executing RELA operations on RS images further amplifies the beneficial impact of the AKT term on image feature vectors and aids in learning more flexible feature representations. The AKT term introduces greater flexibility into the optimization mechanism. This not only enhances positive effects but also establishes new bridges between modalities. This approach provides an explanation for the additional performance improvement observed with the application of the AKT term through more effective cross-modal interaction. These improvements enable ALAC to acquire more comprehensive information and learn more robust feature representations. In addition, after postprocessing with BSWFR, the performance of the model performance further improves by 0.45%. Overall, it is observed that all the proposed learning modules contribute to the enhancement of performance.

To validate the effectiveness of knowledge transfer methods, a series of experiments are conducted to compare various approaches, including smooth $L1$ [51], DIAG [52], SVD [53], and AKT. These methods are applied to two similarity matrices, and their performance is evaluated. The experimental results are presented in Table VI. While some improvements are observed with other knowledge transfer methods to a certain extent, it is evident that our proposed AKT method achieves a 0.36% enhancement in the mR metric compared with the best-performing SVD method. Other methods merely transfer aligned cross-modal information rigidly, which does not enable the model to adapt well to unknown information. In contrast, our approach, constrained by adaptively learned mask matrices, facilitates the reinforcement of positive knowledge transfer and the suppression of erroneous correlations during training, allowing the model to autonomously perceive the direction of information transfer.

TABLE V
EXPERIMENTS WITH DIFFERENT VALUES OF THE PARAMETER λ ON THE RSITMD TEST DATASET

λ	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0.1	6.63	22.34	32.96	**4.75**	**20.39**	**35.75**	**20.48**
0.2	6.45	21.46	32.62	4.22	20.28	35.17	20.27
0.3	6.30	22.34	32.39	4.04	19.73	33.67	20.34
0.4	**6.96**	22.01	32.56	4.44	19.77	35.30	20.14
0.5	6.07	21.57	30.73	4.57	19.35	35.37	20.01
0.6	5.64	22.46	32.63	4.62	19.50	35.46	20.05
0.7	6.30	20.46	33.18	4.22	18.27	32.72	19.19
0.8	6.52	22.01	**34.95**	4.02	19.75	34.22	20.25
0.9	6.41	**22.89**	32.06	4.53	20.26	35.08	20.44
1.0	5.08	20.02	32.52	3.80	18.27	33.14	18.81

TABLE VI
EXPERIMENTS OF DIFFERENT KNOWLEDGE TRANSFER METHODS ON THE RSITMD TEST DATASET

Transfer Method	Audio Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
L1	6.30	20.13	**33.96**	4.18	18.93	32.72	19.37
DIAG	6.41	20.46	32.96	4.57	19.51	34.75	19.78
SVD	**6.85**	21.68	33.18	4.31	20.17	34.53	20.12
AKT	6.63	**22.34**	32.96	**4.75**	**20.39**	**35.75**	**20.48**

E. Parameter Analysis

To further investigate the impact of the parameter λ on the results, we conduct experiments on the RSITMD image–audio dataset. From Table V, it can be observed that when the parameter λ is set to 0.1, our model does not achieve optimal performance in the *R@1*, *R@5*, and *R@10* metrics for audio retrieval. However, in the case of image retrieval, all the metrics reach their peak levels. Furthermore, the overall mR metric for the entire model indicates that this setting achieves the best performance in overall retrieval. In summary, setting the parameter λ to 0.1 allows the model to maintain high recall while exhibiting superior performance in overall retrieval. The experimental results collectively suggest that this configuration is effective in optimizing the tradeoff between audio and image retrieval tasks on the RSITMD image–audio dataset.

F. Performance Visualization of RELA

In this section, we analyze the performance of our proposed method RELA in capturing salient regions in images through visualizing typical experimental results. In Fig. 4, we present the feature extraction results of RELA for different regions.

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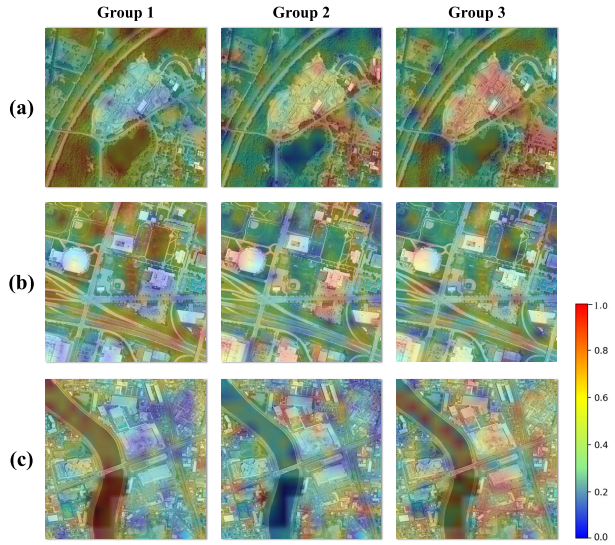


Fig. 4. Visualization comparison of different convolution groups. (a)–(c) Three examples. Darker shades of red indicate higher levels of attention, while darker shades of blue indicate lower levels of attention.

In Fig. 4(a), the image contains salient information such as roads and bare land, and we observe that the three groups of attentions in RELA focus on different regions. In Fig. 4(b), the first group of features mainly emphasizes roads and grasslands, the second group highlights white buildings, and the third group focuses on the regions between buildings and roads. In Fig. 4(c), the first group of attention is concentrated on the river, the second group tends to focus on buildings near the river, and the third group attends to the regions near the roads. From the heat maps on the right side, we observe that our method effectively extracts salient information from different channels, while suppressing redundant information from other channels. This allows our model to attend to different information in different channels, leading to a more comprehensive and accurate understanding of the image. These experimental results further validate the excellent performance of RELA in capturing salient regions in images.

G. Performance Comparison of Different Reranking Methods

In this study, we validate the effectiveness and stability of our proposed BSWFR method by comparing different ranking methods and parameter settings on the RSITMD dataset. In Table IV, we compare the average recall of source retrieval methods [46], rerank methods [27], multivariate rerank methods [28], and BSWFR methods across multiple similarity matrices. In addition, we investigate various methods and parameter values using Figs. 5 and 6 and find that our proposed BSWFR method achieves better results. While the rerank method considers the reverse ranking as the selection of corresponding results, it ignores most of the information from the similarity matrix. The multivariate rerank method averages over multiple similarity matrices, considering both forward and reverse rankings. However, since the similarity matrix itself represents the forward ranking, adding information from the reverse ranking does not express the reverse ranking response well. In contrast, our BSWFR method uses both forward and reverse ranking information as weights to influence the similarity matrix, enabling a deeper exploration of information. The experimental results further support our claim that our method improves retrieval accuracy by 0.44% points on the RSITMD dataset.

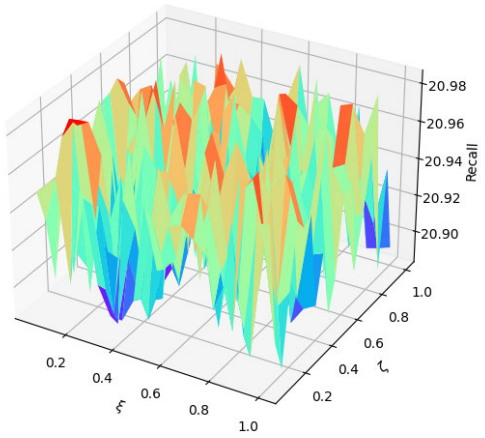


Fig. 5. Adjusting the sensitivity of ranking position quantization with ζ and ξ values affects the reranking results on the RSITMD test dataset.

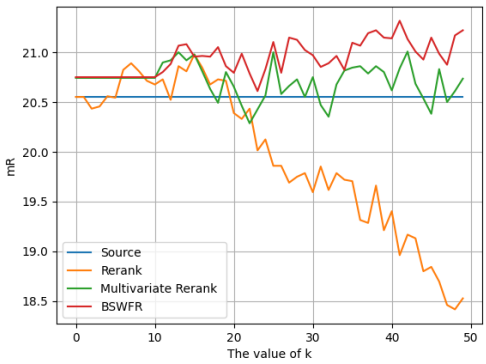


Fig. 6. Cross-modal RSIA retrieval results are obtained on the RSITMD test set using different values of k with various reranking methods. k represents the top k nearest neighbors.

To investigate the impact of different postprocessing methods on the recall of the first k results, we examine the influence of different k values on various approaches, as illustrated in Fig. 6. The results indicate that the reranking algorithm performs well when k falls within the range of 7–20. However, as k continues to increase, there is a sharp decline in accuracy. The multivariate reranking approach achieves favorable results when k is between 21 and 30, but retrieval performance decreases with further increases in k, primarily due to the diminishing effect of reverse ranking. The BSWFR metric exhibits significant fluctuations, but overall, it outperforms the reranking and multivariate reranking methods, demonstrating its robustness.

Fig. 5 illustrates the effect of different ξ and ζ values in (11) and (13) on the retrieval results. They control the sensitivity of the ranking position quantization. The BSWFR algorithm achieves better retrieval accuracy when k is 25, and ζ and ξ are fixed at 0.6–1.25, respectively. The value of ζ has a greater influence on retrieval accuracy compared to ξ, and ξ fluctuates within 0.2 for mR.

Overall, our proposed BSWFR method exhibits better retrieval performance under various experimental conditions on the RSITMD dataset.

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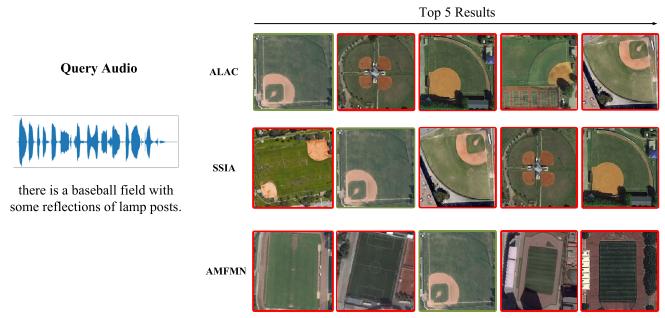


Fig. 7. Top-5 retrieval images for the given query audio on the three methods. Correct retrieval images are marked in green, while incorrect retrieval images are marked in red.

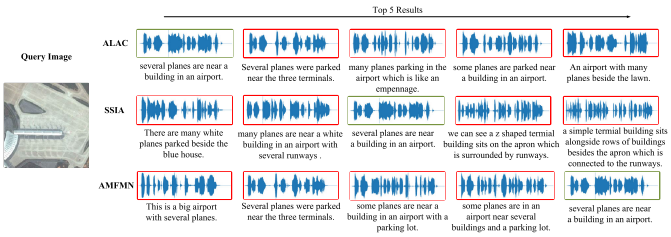


Fig. 8. Top-5 retrieval audios for the given query image on the three methods. Correct retrieval audios are marked in green, while incorrect retrieval audios are marked in red.

H. Visualized Results

In this experimental analysis, we use visualization techniques to assess the performance disparities among various retrieval methods and evaluate their retrieval effectiveness. For a fair comparison, we select two different retrieval methods: the SSIA model, which has a strong ability to remove redundant information, and the AMFMN model, which has a strong ability to extract global features. We compare their top five retrieval accuracies without using any postprocessing operations. From the visual results in Figs. 7 and 8, it can be observed that the retrieval performance of AMFMN is poorer, which is also reflected in the quantitative metrics. In the visual representation of audio retrieval results, as shown in Fig. 7, we can observe that the SSIA model is directly influenced by the input audio content, leading to the suppression of certain image details. This suppression results in some smaller objects in Fig. 7 appearing less prominent. For example, the ground truth of the SSIA model ranks second in the retrieval results, and the first image contains multiple baseball fields, indicating that SSIA focuses more on the information related to baseball fields. On the other hand, the AMFMN model conducts retrieval directly using global features, which results in relatively minor distinctions between candidate instances. The primary information in this audio retrieval task involves baseball fields and the reflection of lamp posts. However, the first two results from AMFMN do not align with the audio description. This suggests that AMFMN, in its feature extraction process, overly emphasizes the most prominent regions in the image, leading to the grouping of different scenes, such as baseball fields and soccer fields, into a single category. In contrast, ALAC mainly uses the RELA module to learn the salient regions of the image, allowing each salient region to be noticed. From the results of ALAC, it can be seen that our model can achieve GT as the first ranking.

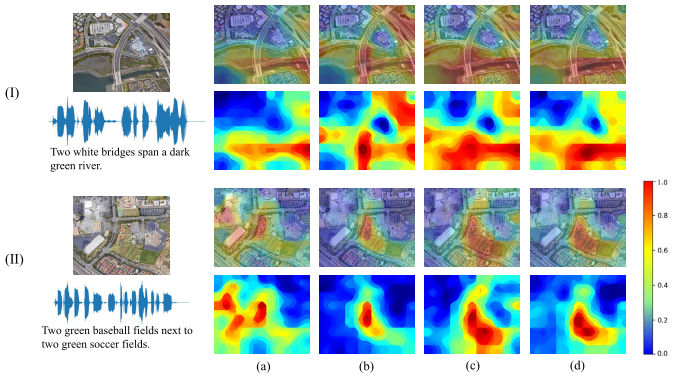


Fig. 9. Visualization of the prominent position between the audio (as a query) and the RS image (as a value). In the second column, the textual description of the content corresponding to the audio is shown. (a)–(d) Visible results of the different modules for the remotely sensed images. (above) Superimposed images. (bottom) Highlighted masks. Darker shades of red indicate higher levels of attention, while darker shades of blue indicate lower levels of attention.

In the results of Fig. 8, when inputting an image of an airport scene, ALAC can retrieve the correct audio description very accurately, while SSIA and AMFMN have more incorrect retrieval results. It can be observed that SSIA is less sensitive to certain color and quantity terms, while AMFMN ignores some local features due to its powerful global feature extraction ability, making it difficult to obtain better visual representation. In summary, based on the above analysis of the results, we conclude that the ALAC approach can better find correlations between images and audio, by paying sufficient attention to the finer-grained information in the image, and thereby accommodating complex and diverse audio descriptions, leading to superior performance in the audio retrieval image task.

I. Semantic Location Visualization

In Fig. 9, we visualize the semantic localization operations for different modules. Fig. 9(a) represents the semantic localization results for the baseline model. Fig. 9(b) represents the semantic localization result with the RELA module removed. Fig. 9(c) represents the semantic localization result with the AKT term removed. Fig. 9(d) represents the effect common to all our modules.

In group (I) of results, the input query audio is “Two white bridges span a dark green river.” As can be seen from the results of our heatmap on the right, the effect in the baseline model can be relatively well-localized to the key information of the river, but there is still not enough attention to the bridge. In group (I)(b), we can see that the model lacks the ability to learn fine-grained regions without the RELA module. The model is able to focus well on the bridge as key information, but the model gives the same attention to the bridge in the other regions that do not fit the audio description. In group (I)(c), we can see that without the AKT term, the model loses the constraint of accurate alignment of different modalities, and the model expands the localization of key information to improve the accuracy of localization, which leads to a large number of features that do not meet the audio description requirements into our feature vector. In contrast, in group (I)(d), our final model has a good

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improvement in both the accuracy of localization information and the range of localization information. And in group (II) of results, we input the query audio information as "Two green baseball fields next to two green soccer fields." The challenge of this image is that there are four soccer fields close to the baseball field, and this is more demanding for the ability of our model to understand some quantitative words. In group (II)(a), the model is unable to locate all the information and pays attention to some information that does not appear in the audio description. In group (II)(b), the model can locate the baseball field and gives partial attention to the soccer field. In group (II)(c), our model can locate the baseball field and the soccer field very well, but the location does not match the audio description. In group (II)(d), we can see that the model can locate the location of the objects well and also meet the semantic requirements of the audio. The above analysis is sufficient to reflect the performance of the proposed ALAC model in semantic localization and cross-modal retrieval tasks.

V. Conclusion

In this study, a novel cross-modal RSIA retrieval method is proposed. This is achieved by incorporating an RELA module to extract distinct salient objects from RS images, thereby enhancing the overall comprehension of images of the model. To guide the learning of aligned feature vectors, a loss function based on AKT is being introduced. Furthermore, to fully exploit the mutual information between different modalities, a plug-and-play rerank approach is being devised, which requires no additional training. Through extensive experimentation, the outstanding performance of our proposed method is being validated across four commonly used RSIA datasets. Compared with the state-of-the-art methods, improvements of 1.49%, 2.25%, 4.24%, and 1.33% on average are being achieved by ALAC. Substantial support is being provided for the realization of cross-modal image–audio retrieval tasks, resulting in significant performance enhancements.

In this study, efforts are being made to enhance the interaction of information between different modalities to address the challenge of information imbalance and ensure consistent representation. However, it is also being acknowledged that the current methods are unable to sufficiently sensitize the audio extraction network to similar word groups. Therefore, future research directions may involve exploring more in-depth approaches, such as segregating the audio of similar word groups, to enable the model to better learn and distinguish them, thereby enhancing the discriminative ability of audio features. In addition, plans are in place to conduct more in-depth investigations in challenging scenarios, such as novel description discovery or weakly supervised cross-modal matching/fusion.

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